



Advanced Physics - Probable Questions

PROBABLE QUESTIONS: (QUANTUM MECHANICS)

1. (a)^{'89} Explain the meaning of the word uncertainty for an observable in Quantum mechanics. Use uncertainty relation to justify that electron cannot be localized within the nucleus.
(b)^{'91} Discuss the uncertainty relation and apply it to some physical problems.
2. (a)^{'91} Establish time independent perturbation theory for a non-degenerate case and apply it to find the eigen values of an harmonic oscillator.
(b)^{'91, '92} Deduce Fermi's golden rule with the help of time dependent perturbation theory.
(c)^{'90} State and explain Rayleigh-Ritz variation principle & use it to obtain an upper bound for the energy of the ground state of Helium atom.
(d)^{'88} Develop the time independent perturbation theory to obtain the 2nd order correction to the energy of a nondegenerate state. Elaborate the condition for the validity of the theory. Discuss these corrections can never be $\oplus$ ve for all ground state.
- 3.^{'91} Apply the Born approximation to find the scattering amplitude and scattering cross-section in the case of elastic scattering. Discuss the validity of this approximation. + Rutherford Scattering formula.
4. (a). Solve Dirac's relativistic wave eq² for a free electron to obtain its energy eigen values. What is the significance of negative energy solution.
(b). Establish the relativistic invariance of Dirac eq².
(c). Show that the law of conservation of the angular momentum leads to the result that Dirac eq² describes a spin- $\frac{1}{2}$ particles.
(d). Reduce the Dirac eq² in the presence of the e.m. field.



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On the non-relativistic form under weak field approximation (or alternately in the presence of the magnetic field) and establish the existence of the spin magnetic moment of the Dirac particle.

(5) Using the theory of Heitler and London, find out the energy in the normal state of the hydrogen molecule. Discuss your result.

(6) Electron Spin & Pauli theory, ^{Simple application} to an atom in mag. field

Symmetric and asymmetric wave functions Pauli's principle

MATRIX MECHANICS

(6). Solution of Simple harmonic oscillator problem.



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Guess Questions for Msc. (Long ans):

Solid State physics:

Super conductivity:

- ✓ (1) Give qualitative idea of BCS theory of Superconductors. What is B.C.S Ground State Distinguish between energy gap of a superconductor & band gap of insulator.
- ✓ (2) Discuss the nature of energy gap in a Superconductor. How does its width depends on temperature. Give an account of D.C Josephson effect.

Magnetism:

- ✓ (3) Discuss the theory of antiferromagnetism using the two sublattice model. Find the ratio of χ and explain how the susceptibility varies with temp. T_N below the Neel temperature T_N .
- ✓ (4) Give Quantum theory of ferromagnetism. Define Curie temp. and show that magnetism of ferromagnetism material arises due to electron spin rather than due to orbital motion.
- ✓ (5) Give the various types of magnetic Ordering & give the elementary theory of antiferromagnetism.

Electron States:

- ✓ (6) Describe the nearly free electron model of a solid. Obtain approximate solution of wave equation near a zone boundary and hence discuss the occurrence of energy band gap.
- ✓ (7) Describe the nearly free electron models for determining the electron energy bands in solids & show that the model deals with finite discontinuity in energy the Brillouin zone boundary.



- (8). State and prove Bloch's theorem for elastic wave passing through periodic structure. Explain how it helps in underlining the properties of solid.
free electron Gas in 3D periodic potential.
TIGHT BINDING METHOD.

(9). Derive Laue eqⁿ for diffraction of X-rays from crystals & hence obtain Bragg's law. Note on Reciprocal Lattice & properties.

(10). State and prove Bloch theorem. Discuss the value of k involved for a crystal. Show that the wave function:

$$\Psi_k(r) = \sum_{i=1}^N e^{ikR_i} \phi(r-R_i) \quad \text{Can be expressed as a Bloch function.}$$

(11). Explain the tight binding method for obtaining the wave function of electron in solid use it to obtain one electron energies for b.c.c & f.c.c crystal.

+ Defects in crystal & all short questions.
+ previous years Qs. Note.

Guess Questions for MSc (Nuclear physics):

- (1). γ -ray interaction with matter: Compton Scattering, Pair production, photo electric absorption. Describe origin of γ -rays. Specify the energy region in which each of the processes has a dominant concentration.
- (2). What are the experimental evidences in support of the shell structure in nuclei? Describe



- 11a). how the magic numbers may be explained on the basis of shell model employing a central potential with spin-orbit interaction.
- ④ GROUND STATE OF DEUTRON NUCLEUS.
- ⑤ Give the theory of low energy n-p scattering. Obtain an expression for the elastic cross-section in terms of phase-shifts.
- ⑥ write down a relativistic eqⁿ for π -mesons and then show how Yukawa potential can be obtained from that eqⁿ. Estimate the mass of the field carrying mesons. If the range of Nuclear force is 2×10^{-15} m.
- ⑦ Define "Nuclear Reaction Cross-Section" & "Differential cross-section". Obtain an expression for the cross-section of a nuclear reaction initiated by neutrons of orbital angular momentum l .
- ⑧ Describe an accurate method for the determination of the magnetic moment of proton. what information is obtained about the structure of nuclei from the knowledge of their magnetic moments.
- ⑨ COSMIC RAYS.
- ⑩ Principle of phase stability in Synchrotron Betatron. Describe Proton Synchrotron, How is strong focusing achieved in proton synchrotron?
- ⑪ Elementary particles. Quarks: Strangeness.



- ⑩/ what are Spontaneous fission and Induced fission? Describe briefly how the fission phenomenon can be explained on the basis of liquid drop model.
 - ⑪/ Obtain a formula for the loss of energy suffered by a charged particle in passage through matter. Hence derive an expression for the range of the particle in the medium.
 - ⑫/ Breit - Wigner Dispersion formula for $l=0$ neutron
 - ⑬/ Semi-empirical mass formula. Parity.
- + all Short Questions & previous years Qs. NOTE



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Probable Questions : (General Electronics) :

SPL DIGITAL circuit

Q.1. (a). R-C Coupled Amplifier. (Analysis with cut mid-freq)

Q.1. (b). Sketch the circuit of a peak-detector and explain its operation.

Q.1. (c). Discuss the biasing circuit used in BJT. JFET - Comment on the Q-point stability with respect to different biasing.

Feedback Concept

Q.2. (a). What is negative feedback in amplifiers. Discuss the General characteristic of negative feedback circuit. Give the circuit of an emitter follower and discuss its operation.

(b). Deduce the negative feedback equation and discuss the characteristics of a negative feedback circuit. (Nyquist's rule)

Oscillators

3. (a). Discuss the principle, working and design of an R-C phase shift Oscillator.

(b). Crystal Oscillator: Tuned circuit Oscillator.

Modulation:

4. (a) What is amplitude modulation? Describe the plot of frequency spectrum in amplitude modulated wave. Describe with necessary circuit the operation of AM modulator using solid state device.

(b) ¹²F.M modulation (for Short-arc).

DIGITAL ELECTRONICS

5. (a). Discuss with necessary circuit the operation of a transistor logic circuit.

(b). What is a "flip-flop". Discuss the operation of a J-K Master. (Comit-25).



Probable Qns from Mathematical Physics

Qn. (1). Define the Fourier transformation $F(k)$ of a function $f(x)$ and discuss its properties. Obtain the Fourier transformation of Gaussian function. $f(x) = Ne^{-ax^2}$ & discuss the result. a & N are constants.

Qn. (2) (a) Explain what do you understand by Laplace's transformation of the following functions
(b). what is meant by Convolution of two functions $f(t)$ and $g(t)$. State and prove Convolution theorem use it to solve damped harmonic oscillator.

Qn. 3. Green's function: Problems.

Qn. (4). Matrix: Tensors: Gr. Theory.

Electrodynamics and Relativity:

(1) (a) Discuss Lorentz transformations as orthogonal transformations in 4 dimensions. (LIGHT CONE).

(6) Energy momentum Vector/Tensor in electromagnetic field.

(7) (a) Radiation from Oscillating dipole. (b) Total power radiated from an accelerated charge; Larmor formula.

(8) (a) Show that Maxwell's eq^s are covariant under Space time transformation.



Selected Short Qs from General Electronics

- (1) Compare the characteristics of the phase shift and Wein bridge Oscillator. [96 M+H] [G+K Page 53 + 400]
- (2) Explain the action of clipper, biased clipper and combination clipper circuits.
- (3) Show that two emitter coupled BJT's can act as a differential amplifier.
- (4) What is the advantage of frequency modulation over amplitude modulation in signal transmission.
- (5) What is "flip-flop"? Discuss the truth table of a clocked J-K flip-flop.
- (6) What is the relation between "Bode-plot" and "Nyquist plot". [B.G. 505 M+H : 473 + COPY]
- (7) State and prove Miller's theorem. [384 B+]
- (8) What is Karnaugh map? What is its importance in digital electronics. [COPY] [677 B+]
- (9) Why the equivalent input capacitance for a common emitter a.c amplifier is so large. What circuit capacitance dominate amplifier response at low and high frequencies? [378]
- (10) What is a photo-diode give some of its application.
- (11) Explain how R-point stability is achieved. (148 B+)



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- (12) what do you understand by "race condition" in S-R flip-flop. [M4H: 6, 2, 5]
(13) Give the logic diagram of X-OR gate using NAND gates only. [C73]
(14) Explain the use of Zener diode in regulated power supply. [C73]
(15) Explain the effect of negative feedback on the distortion of an amplifier. [407 + COPY]
(16) what is saw tooth wave discuss one of its use. [COPY + 525]
(17) Give a diagram of clipping circuit & point out its use. [COPY + 525]
(18) what is a push-pull circuit? Give its advantage in case of an amplifier. [998]
(19) what do you mean by phase modulation.
(20) what do you mean by current feedback explain with diagram. [412]
(21) what is flip-flop circuit. Explain with diagram.
(22) F.M. Production & detection. [COPY]
(23) Differential amplifier.
(24) Logic gates, Shift registers; Dis. amp. [669]

Selected Qns. from Relativity & Electrodynamics.

- (1) Prove that velocity of light in free space is the highest velocity that a material particle can attain.
(2) Prove that Lorentz transformation is an orthogonal transformation.
(3) what is the Gauge transformation? Discuss briefly Lorentz Gauge.
(4) Prove orthogonality of momentum four vector and acceleration four vector.
(5) Show that a four dimension element of volume is invariant under Lorentz transformation.



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- (6) Explain what you understand by light cone.
- (7) What do you understand by electromagnetic field tensor.
- (8) State and explain uncertainty principle.



① Selected SHORT Questions: 26.2.95

Mathematical Physics:

①. Show that $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$ where x is complex variable.

(2) Define Dirac-Delta function and show that $\int_{-\infty}^{+\infty} f(x) \delta(x-a) dx = f(a)$ symbols have their general meaning. [162: chapt-10].

(3) Calculate Green's function in case of a loaded stretched string satisfying differential eqⁿ
 $\frac{\partial^2 u}{\partial x^2} = f(x)$ [186]

(4) Obtain the Cauchy - Riemann Condition for differential function $w = u + iv$. State and prove Cauchy's theorem for an analytic function.

(5). Show that every function $f(z)$ analytic is an annular $R_1 < |z-a| < R_2$ can be expressed as a Laurent Series.

$$\sum_{n=-\infty}^{+\infty} C_n (z-a)^n; \quad C_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z) dz}{(z-a)^{n+1}}$$

Symbols have their usual meaning.

(6). Define the Fourier transformation $F(k)$ of a function $f(x)$ and discuss its properties. Obtain the Fourier transformation of the Gaussian function [321 chapt-4]
 $f(x) = N e^{-ax^2}$ (N & a are const) and discuss the result.



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- (7). Prove that the Wronskian is either identical zero or never zero.
- (8). Prove that the Delta-function is an even function. [163 chapters]
- (9). Explain what do you understand by Poles and Singularities of a function.
- (10). what is the use of Green's function in mathematical physics? [176] chapters.
- (11). Show that the multiplication of tensors [339] ^{chapters} results in the addition of their ranks.
- (12). Explain what you understand by light cone.
- (13). Explain the use of Fourier-Transform in mathematical physics. [324 chapters].
- (14). State and explain Cauchy Integral theorem.
- (15). what is Dirac-Delta function? Give some of its important properties. [164] chapters.
- (16). what do you mean by Contour integral explain.
- (17). State and explain the properties of Green's function. [178 chapters]
- (18). State and explain Cauchy's residue theorem.
- (19). what do you understand by the electromagnetic field tensors.



II Selected Short Questions from QUANTUM MECHANICS.

(1) Why are dynamical variables represented by linear Hermitian Operators in wave mechanics. [428]

(2) Show that the eigenvalues functions of $\hat{L}^2$ have the same parity as the parity of L .

(3) If the two non-Hermitian operators are defined as below

$$\hat{a} = \left(\frac{m\omega}{2\hbar} \right)^{1/2} \hat{x} + i \left(\frac{1}{2m\hbar\omega} \right)^{1/2} \hat{p}_x, \quad \hat{a}_+ = \left(\frac{m\omega}{2\hbar} \right)^{1/2} \hat{x} - i \left(\frac{1}{2m\hbar\omega} \right)^{1/2} \hat{p}_x$$

find the value of the commutator $[\hat{a}, \hat{a}_+]$.

(4) Show that Second order perturbation correction to energy is necessarily negative if the unperturbed state is the ground state. [344+396]

(5) For the case of elastic scattering derive the relationship between the differential scattering cross-section and the scattering amplitude [286]

(6) Show that the operator $\hat{K}$ defined through $\hat{K} = \beta (\underline{\sigma} \cdot \hat{L} + \hbar)$ commutes with the operator H .

(7) Explain the physical significance of Schrödinger's wave function. [76]



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- (8). Find the operator for the co-ordinate x in momentum representation. [428] [158]
- (9) Find the Operator for L^2 . Identify its eigenfunction and eigenvalues. [608]
- (10). Discuss the significance of Einstein's A and B coefficients. [383] + COPY.
- (11) Discuss the symmetric and anti-symmetric wave function. [496]
- (12). Establish a relation between the scattering cross-section in the laboratory co-ordinates system and that in the centre of mass co-ordinate system. [288]
- (13). Discuss why Klein-Gordon equation could not be applied to a relativistic free electron. [643]
- (14) Distinguish between Schrödinger picture and Heisenberg picture of representation of quantum mechanical system. [553]
- (15) Discuss the time dependent perturbation theory of a quantum mechanical system. [364]
- (16) Explain with examples the difference between induced and spontaneous emission. [383]
- (17). Discuss the validity of Klein-Gordon equation. How it can be used to obtain the probability current density. [642]

Short Qs. from Relativity + Spectroscopy:

- (1). Explain Spin-orbit coupling. [169]
- (2). State and explain Frank-Condon principle in spectroscopy. [COPY] + 350]



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- (3). Write a brief note on microwave Spectroscopy of Gases.
- (4) State Larmor's theorem. Derive an expression for Larmor's frequency.
- (5) What are Russell-Saunders and $\vec{J}-\vec{J}$ coupling schemes? Discuss their suitabilities. [181]
- (6) Illustrate through the example of Sodium-D-lines the difference between Paschen-Back effect and Zeeman effect.
- (7) Distinguish between fine and hyperfine structure in atomic spectra.
- (8) What factors control the width of spectral lines? What is self width?
- (9) Indicate the general features of the electronic spectra of diatomic molecules.
- (10). Calculate Landé g -factor for $^{10}\text{H}_{1/2} \leftarrow ^{10}\text{S}_{1/2}$ levels [200]
- (11). Draw diagrams to illustrate D-lines of Sodium
- (12) Distinguish normal & Anomalous Zeeman effect. [202]
- (13) To what extent the vibrating rotator model agree with expt. result for the spectra of diatomic molecules.



III. Selected Short Qrs from Solid State Physics

- (1) The primitive translation vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ for f.c.c. Structures are all equal in magnitude and angle between any two of them is α . Calculate α .
- (2) Show that the Simple Cubic lattice is Self reciprocal.
- (3) Show that thermal expansion of a crystal necessarily requires consideration of anharmonicity.
- (4) "Diamagnetism is a universal property" Comment on the above statement.
- (5) Show that how the photoconducting property of a solid material is influenced by the presence of electron traps.
- (6) Explain why type (ii) Superconductors are used for the construction of Superconducting magnets.
- (7) Show that f.c.c lattice is the reciprocal lattice of b.c.c lattice.
- (8) How do you predict whether a particular bonding is predominantly metallic or predominantly covalent?
- (9) Discuss the failure of free electron model



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- (10). Distinguish between intrinsic and extrinsic Semi Conductors.
- (11). Explain the origin of permanent magnetic moment.
- (12) Discuss the cause of NMR and some of its applications.
- (13) State some of the properties which do not change in Superconducting Transition.
- (14) Give some comment on London formula.
- (15). Show that reciprocal lattice vector γ_{hkl} is normal to (h, k, l) planes of the crystal lattice.
- (16) Can neutrons of energy 0.08 eV be suitable for diffraction from crystals? $m_n = 1.675 \times 10^{-24} \text{ gm}$
 $h = 6.6 \times 10^{-27} \text{ erg sec.}$
- (17) Show that the K.E of a three dimensional gas of N-free electrons at 0°K is given by $E_0 = \frac{3}{5} N E_F$ where E_F = Fermi Energy.
- (18) Discuss the concept of effective mass of electrons in crystals.
- (19) Explain the origin of thermal expansion.
- (20) What is Meissner effect? Explain the importance of Meissner effect & Isotope effect in the study of superconductivity.



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- (21). Explain the Concept of energy gap in metallic Superconductors and comment on its experimental evidence.
- (22) Explain the meaning of the Spin lattice and Spin relaxation.
why resonance cannot be observed for isolated nucleus?
- (23) why some metals which are very good conductors at normal temperatures do not become Superconductors.
- (24) what are the properties of Superconduct
- (25) Explain: Excitons, Tight binding approximation, Single particle tunneling; Ferrimagnetic order

Short Qs from Statistical Physics:

- (1) Assuming rigid rotating molecule Calculate the (i) rotational partition function & (ii) rotational heat capacity for chlorine gas at 27°C . The m.m. given by $114.5 \times 10^{-27} \text{ gm cm}^2$.
- (2) Elucidate the concept based on statistical mechanics of entropy.
- (3). Discuss Canonical and Grand Canonical ensembles.
- (4) Explain what is meant by Brownian motion & transitional Brownian motion (Langevin)



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- (5) How Bose-Einstein Condensation is different from ordinary Condensation.
- (6) Explain the Concept of fluctuations in thermodynamic quantities.
- (7) Compare with the help of diagrams the temperature dependence of the heat capacity of a gas in three statistics.
- (8) Obtain an expression for rotational partition function of H_2O molecule at high temperature. What are the symmetry numbers for HCl , NH_3 , CH_4 & CH_3Cl molecules?
- (9) Prove that in classical statistical mechanics the equilibrium time correlation function of two dynamical variables is stationary.
- (10) Discuss for Fermi gas $S \rightarrow 0$ when $T \rightarrow 0$ though V and P has finite values.
- (11) "Some of the properties of second virial coefficient at low and high temp are different" Discuss.



IV. Short Questions from NUCLEAR PHYSICS

- (1). There cannot be any nuclear reaction without some scattering although scattering without reaction is possible - Elucidate. [415]
- (2). Why in Compton Scattering would you expect $\Delta\lambda$ to be independent of which scattering is composed.
- (3). Give the qualitative explanation why an α -particle can penetrate Coulomb barrier?
- (4). How do you account for the negative magnetic moment of the neutron? [copy]
- (5). Why is the proton-proton scattering differential cross-section nearly symmetric about 90° in the centre of mass frames of reference.
- (6). Can we say that the $\bar{\pi}^0$ -meson is its antiparticle? Do all particles have antiparticle? What about photon? [copy]
- (7). What are the four fundamental interactions of nature? Mention the field quantum for each of them?
- (8). What are hyperons? Complete the following decay schemes [copy]
 - (a) $\eta \rightarrow p + \bar{p} +$
 - (b) $\Sigma^0 \rightarrow \pi^0 +$
 - (c) $\Xi^{--} \rightarrow + \pi^-$



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(9). Mention the high resolution solid state detectors that are used for the studies of low energy β - γ transition of nucleides. why can not the radiation of multipole order 0 occur.

(10). NOTE

(a) Cosmic ray Shower

(b) Non conservation of parity in β -decay.

(c) Quarks

(d) Semi empirical mass formula

(e) Betatron.

(f) Positronium.

(h) The Compton effect.

(i) Cerenkov detector

(j) Origin of Cosmic rays.

(k) Gellman Nishijima scheme

(l) Nuclear fusion.

(m) Linear accelerator.

(n) Isotopic Spin & Strangeness.

(o) Mossbauer effect.

(11). Explain the physical meaning of non-conservation of parity in weak interaction. [60%]

(12) State why no γ -radiation of multipole order can occur although emission of internal conversion electron is possible [60%]

(13) why NaI(Tl) Scintillator is considered as an efficient γ -detector? [60%]



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- (14). write down the basic quantum numbers associated with an elementary particle.
- (15) what are different types of interaction of particles? Mention the quanta carrying the fields.
- (16). How do you account that π -mesons are iso-spin triplet.
- (17) why it is believed that repulsive core of the nucleon potential arises from the exchange of mesons heavier than the π .
- (18) Explain the physical significance of the (-)ve magnetic moment of neutron. [100]
- (19) what reasons have you to believe that there are two kinds of neutrinos. [100]
- (20). Suggest some nuclear models that explain giant resonances.
- (21) why is it believed that the nuclear forces saturates? [100]
- (22) Complete the following decay schemes:
- $$p \longrightarrow \eta + \beta^+ +$$
- $$\pi^- \longrightarrow \quad + \gamma^-$$
- $$\Lambda \longrightarrow p +$$



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- (23) Why is the P-P Scattering differential cross-section nearly symmetric about 90° in C.M frame of reference.
- (24) Describe a high resolution Semiconductor detector which is used in the field of low energy X-ray Spectroscopy.
- (25) How do neutinos differ from photons which also have no charge or rest mass?
- (26) Is there a physical picture of how the momentum of a π -meson transferred between the fields of two nucleons leads to an attractive force between them.
- (27) Explain why there can be no collisions between a typical nucleon and another in a nucleus in its ground state.
- (28) Why light nuclei are more effective moderators than heavy nuclei?
- (29) Explain the critical size of thermal reactor.
- (30) Compare P-P & n-p Scattering.
- (31) What do you understand by allowed & forbidden transitions?
- (32) Validity of Bohr's compl. nuclear theory.
- (33) Why slow neutron can induce fission.



(34) Does parity conserved in all interaction
[COPY]

(35). Explain that a free electron cannot absorb a photon & conserve both energy and momentum in the process.

(36). Why ^{235}U is more fissionable than ^{238}U ?

(37). what are the Selection rules for the multipole γ -ray transitions?

(38) How do the nuclear forces differ from the forces involved in weak and electro-magnetic interactions?

(39) what information do the n-p and p-p scattering provide regarding nuclear force

(40) Explain that the decay

$^{14}\text{O} \rightarrow ^{14}\text{N}^* + \beta^+ + \gamma$ is a pure Fermi transition where as

$^6\text{He} \rightarrow ^6\text{Li} + (\beta^- + 2\gamma^-)$ is a pure Gamow-Teller transition.

(41) what are the reasons to believe that the ground state wave function of the deuteron is predominantly in the 3S_1 state with a small admixture of 3D_1 ?